# 9 Static Methods




## KEY IDEAS

- `@staticmethod` is a plain function that lives in the class namespace for organisation. It gets **no** `self` **or** `cls`, so it cannot touch instance or class state.
- Callable on the class or on an instance. Good for helpers, validators and conversions that belong with the class.
- Need class data or must build instances? Use `@classmethod` instead.

## EXAMPLE

```python
class Temperature:
    @staticmethod
    def c_to_f(c):                    # no self, no cls
        return c * 9 / 5 + 32

    @staticmethod
    def is_valid(c):
        return c >= -273.15

    def __init__(self, c):
        if not self.is_valid(c):      # via self or class
            raise ValueError("too cold")
        self.c = c

print(Temperature.c_to_f(100))        # → 212.0
print(Temperature(20).c_to_f(0))      # → 32.0
print(Temperature.is_valid(-300))     # → False
```